DIGITAL ASSET INVESTMENT FEATURES

Learning Outcome

- explain investment features of digital assets and contrast them with other asset classes

Digital assets have a growing presence in the financial services industry and have become a more prevalent asset class for investors. Most digital assets are cryptocurrencies, and their growth has been spectacular. While there were around 70 cryptocurrencies recorded in 2013, by early 2022, there were close to 10,000 different cryptocurrencies issued by corporations, organizations, and in many cases, individuals.

Cryptocurrency investment is typically viewed as an alternative asset. As digital assets increase in prominence, some institutional investors selectively seek exposure to these assets for their potential higher returns and possible diversification benefits. When institutional investors become more comfortable with investing in digital assets, it may be a very early indication that the market may be close to reaching a tipping point. As more institutional investors are becoming comfortable with the idea of owning digital assets outright, various financial service providers, such as digital exchanges and custodians, have expanded their infrastructure in anticipation to support potential investment in these digital assets.

Distinguishing Characteristics of Digital Assets

There are notable differences between digital assets and traditional financial assets. The main similarity between these asset types is the emergence of indirect investment vehicles such as exchange-traded funds and hedge funds that invest both in traditional financial assets and in digital assets. The main differences between these asset types are as follows:

- *Differences in inherent value*. Unlike financial assets, most digital assets do not have an inherent value based on underlying assets or on the potential cash flow—interest and dividends—they can or are expected to generate. With no expected earnings to inform today's value and indicate the likely direction for tomorrow's prices, digital assets do not have a fundamental value. They derive their price solely from an anticipated asset appreciation from a perceived scarcity value (due to limitations on the total supply of currency) and the potential ability to transfer value in the future (due to unique features in the underlying algorithms that may facilitate certain types of financial transactions).
- *Differences in validating transactions.* One key difference between digital assets and traditional financial instruments is that traditional assets are generally recorded in private ledgers maintained by central intermediaries. The ownership and exchange of digital assets are generally recorded on a decentralized digital ledger using cryptography and enhanced algorithms. Whether a digital asset transaction is validated in a permissionless or permissioned network using either a PoS or PoW standard impacts its perceived value.
- *Differences in the uses as a medium of exchange*. Financial assets are traditionally priced and traded in widely accepted currencies; they can be readily transacted and exchanged into fiat currencies. Fiat currencies form the foundation of modern financial systems as the mediums of exchange in any country. In contrast, some digital assets—for example, cryptocurrencies, such as Bitcoins—are used as direct substitutes of real-world fiat currencies in certain circumstances—especially for online transactions in the emerging Web3 ecosystem. Web3 is a concept that refers to the third iteration of the

- While many digital assets, such as digital art and NFTs, are transacted and traded using cryptocurrencies, there is only a very limited acceptance of digital assets as a medium of exchange in the mainstream financial system to date. A reason has been the relative cost of the transactions, which can be prohibitively high to meaningfully process a large volume of transactions. While cryptocurrencies are being used to pay for certain goods and services and as investments, they are not legal tender in most jurisdictions and cannot usually be used to legally settle and extinguish debt in the same way as fiat currencies. Finally, the use of cryptocurrencies for payment and investment purposes is severely restricted in multiple countries; some countries have banned its usage outright and digital asset ownership may result in criminal conviction. Some governments are exploring the concept of CBDC as a tokenized version of fiat currencies, but the jury is still out how the idea will evolve.

EXAMPLE 1

Tesla's Acceptance of Bitcoin and Dogecoin

A high-profile supporter of cryptocurrencies and tokens has been Elon Musk, the CEO and major shareholder of Tesla Inc., the electric vehicle automaker. In February 2021, Tesla announced in a public filing that it has invested USD1.5 billion in Bitcoin to diversify its cash balance (of over USD19 billion at the end of 2020). It also indicated it would start accepting Bitcoin as a payment method for its vehicles subject to applicable laws and on a limited basis initially. But in May 2021, the company announced that it had suspended vehicle purchases using Bitcoin, citing concerns about increasing use of fossil fuels for Bitcoin mining and transactions. In July 2021, Musk indicated that Tesla would most likely restart accepting Bitcoin as payments once it had conducted due diligence on the amount of renewable energy used in Bitcoin mining. It resumed accepting Bitcoin; in January 2022, Musk announced that Tesla would start accepting Dogecoin as a payment method for selected brand merchandise. It is now the only cryptocurrency that Tesla accepts for online payment for certain Dogecoin-eligible products.

In July 2022, Tesla announced that it had sold off 75% of its Bitcoin holding as the price had plummeted, but none of its Dogecoin.

- *Differences in legal and regulatory protection*. The rules governing financial instruments and the trading of financial instruments are clear, predictable, and well defined across most jurisdictions. Legal and regulatory standards for traditional financial instruments and assets and trading in these are well developed and are often comparable across national borders; this is not the case with digital assets. Comprehensive rules that are specific to digital assets—their formation, trading, and legal standards—are still being developed. The unclear, ambiguous framework puts digital assets at a disadvantage. US regulators consider certain digital currencies as commodities, and other regulators consider them a non-financial asset. With no comprehensive and clear legal recognition—and protection—for digital assets, their value is highly speculative. Finally, the various exchanges at which digital currencies are traded are typically not regulated as a traditional financial exchange. Therefore, certain market behaviors, such as price manipulation and outright fraud, that are considered to be illegal activities on traditional financial transactions are not necessarily prohibited on a digital exchange. Exhibit 5 shows differences between digital and traditional financial assets.

Exhibit 5:

	Digital Assets	Traditional Financial Assets
Inherent value	▪ No fundamental value or future cash flow generation ▪ Price driven by certain features on the blockchain	▪ Value determined by future cash flow generated from the assets
Transaction validation	▪ Usually recorded on decentralized digital ledgers using cryptography and algorithms for permissionless blockchain networks	▪ Recorded in private ledgers maintained by central intermediaries
Uses as a medium of exchange	▪ Very few digital assets are used as a direct medium of exchange, mainly targeting large-scale commercially viable acceptance	▪ Not used directly as a medium of exchange but can be readily transacted and exchanged into traditional fiat currencies that are widely used in the real world
Legal and regulatory protection	▪ Ambiguous, often contradictory, evolving framework; generally unregulated, with minimal legal protections ▪ Use can be illegal or criminal in some countries	▪ Well-established, tested, and proven legal, regulatory, and commercial standards that are clear, predictable, and well defined across all jurisdictions

Investible Digital Assets

The rapid growth of digital assets has been remarkable. That several digital assets and cryptocurrencies exist reflects that their underlying blockchains are designed and optimized for different purposes. The most popular cryptocurrencies are Bitcoin and Ether. Bitcoin and Ether together make up over 80% of total cryptocurrency market value as of July 2022.

Bitcoin (BTC or XBT) and was launched to secure payments in a P2P network. It was designed as an alternative to traditional currencies: as a medium of exchange and store of value. It is the best-known and most widely traded cryptocurrency and its fundamental design continues to influence the development of new types of digital assets. Thousands of other cryptocurrencies exist based on technology similar to Bitcoin: These are called **altcoins**.

Altcoins

The most prominent altcoin is **Ether**, launched in 2015 on its own Ethereum network. Ether involves the added feature of a programmable blockchain that allows users to construct applications using the blockchain to validate or secure transactions or payments for markets or other uses. Ether and other programmable coins therefore have the possibility to be more widely used than simply as a store of value. Such programmable altcoins are also called smart coins or smart contracts, a self-executing contract with the salient terms of the contract directly written into the lines of code, which allows for execution of the contract using the blockchain network. That execution provides a trackable and irreversible record, subject to immutable verification by the nodes of the network. Other altcoins include stablecoins and meme coins.

Stablecoins

Stablecoins are designed to maintain a stable value by linking their value to another asset and are collateralized by a basket of assets, typically legal tender, precious metals, or other cryptocurrencies. The reserve basket protects the holders from price volatility and

algorithmic stablecoins—are designed to use algorithms to control the available supply of the asset, such as minting additional assets when there is increased demand for the coin. It is important to note that stablecoins cannot be exchanged for fiat money and do not have any legal or regulatory backing.

Stablecoins may potentially facilitate settlement and streamline cross-border trading, investing, and payments. In recent years, several competing solutions have been developed to facilitate transactions across various physical and tokenized financial assets and instruments using the stablecoin concept. A special example is the **asset-backed token**, which maintains price parity with some target asset (the US dollar or gold) through tokenization and will be covered later in this module. For an investor holding assets in in a cryptographic wallet, asset-backed tokens provide exposure both to assets that are on the chain (cryptocurrencies and digital assets) and off the chain (financial instruments).

EXAMPLE 2

Market Shakeout of Stablecoins

Stablecoins are designed to track the value of real-world conventional fiat currencies and to provide immediate access to liquidity for traders to redeem their digital stablecoins at par with the value of fiat currencies. Some of these stablecoins are collateralized by financial assets, such as a basket of fiat currencies, gold, Treasuries, and other liquid or illiquid assets. Two of the largest stablecoins in terms of collateral, Tether and USD Coins, use this approach and can either be traded freely on the open market or be redeemed for their dollar value from the issuers.

Another popular stablecoin, Terra, was often called an “algorithmic stablecoin.” It was backed not by collateral of financial and physical assets but through an algorithm that sought to keep its value stable.

For instance, TerraUSD was Terra’s stablecoin pegged to the US dollar “algorithmically” instead of the value of the underlying collateral in US dollar reserves. The computer algorithm developed by the issuer of Terra, Terraform Labs, used Luna, a governance token also issued by Terraform Labs, to maintain the stability of Terra’s value. The relationship between Terra and Luna was designed so that the holders of Terra could always redeem it for a stable value of newly minted Luna coins. If the pricing relationship between Terra and Luna was not in equilibrium, new Luna coins could be issued or existing Luna coins would be destroyed to maintain the value relationship and this algorithmic peg between Terra and Luna. Through financial arbitrage, the stability of Terra’s value by trading between Terra and Luna would be held. Effectively, Terra was simply backed by Luna, both bearing a US dollar value.

In May 2022, this relationship was tested. On 9 May, the price of Luna first began to slide, putting pressure on the Terra’s peg. Many Terra owners rushed to redeem their coins, leading to a sudden surge in supply of newly issued Luna: In just one week, the number of Luna increased from 350 million coins to 6.5 trillion. Consequently, the price of Luna collapsed, as did that of Terra.

TerraUSD was finally depegged from the US dollar on 9 May 2022. Though Terra’s co-founder tried to revive Terra in a reincarnated form of TerraClassicUSD, the stablecoin had since lost its credibility. With its appeal diminished, the coin was practically defunct.

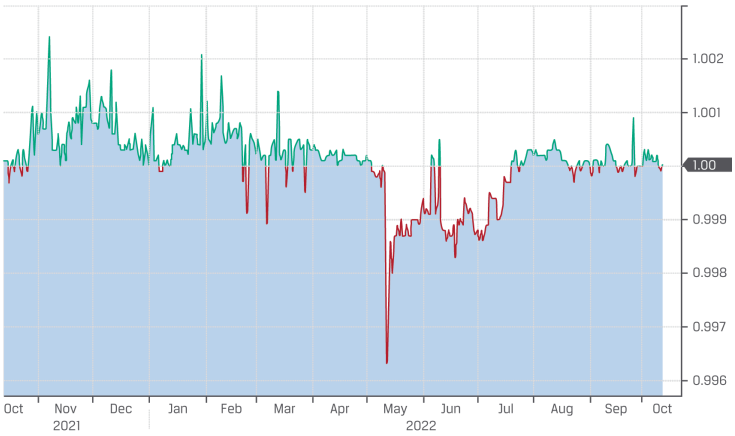
The collapse of Terra sparked serious concern over the collateral backing of its largest stablecoin counterparts, Tether and USD Coins, because of the limited transparency of their reserves. A massive run and redemption by nervous holders of these stablecoins soon followed.

While the “securitized” stablecoins, such as Tether, may have survived market turbulence, other “algorithmic” stablecoins may not when their technical flaws are

shown in Exhibit 6.

Exhibit 6:

Market Price of Tether



Source: Bloomberg

Meme Coins

Meme coins are cryptocurrencies often inspired by a joke and are generally launched for entertainment purposes as they gain popularity in a short period of time, which allows early purchasers to sell their holdings at an often considerable profit.

EXAMPLE 3

Meme Coins—The History of Dogecoin

The original and probably most popular meme coin is Dogecoin, which is based on the popular "doge" internet meme. Dogecoin, featuring a Shiba Inu dog on its logo, was originally founded in 2013 as a mockery of the popularity of cryptocurrencies, mainly Bitcoin. Even the name "dog" was misspelled as "doge" as part of a light-hearted parody.

While some cryptocurrencies are scarce by design, Dogecoins were intentionally designed to have no limit in supply. But what started as a joke soon gained popularity among fervent followers as a cult symbol, especially after social media endorsements by some high-profile business figures. At its peak in May 2021, Dogecoin's market value had jumped to over USD80 billion. However, the price and market cap of Dogecoins plummeted as investors and speculators dumped these coins because their value seemed hardly justifiable. By May 2022, its market dropped to USD11 billion.

Question Set

1. The market value of a digital asset is primarily driven by:

A. the future price expectation of speculators.
B. the future earnings generated from the digital assets.
C. the cryptographic algorithm of the blockchain network.

Hide Solution

The correct answer is A. Unlike financial assets, most digital assets do not have an inherent value based on underlying assets or on the expected cash flow that can be generated. In other words, digital assets do not have a fundamental economic value. Their prices depend solely on the expected asset appreciation due to the perceived scarcity value and the potential ability to transfer value in the future; however, that value reflects market expectations at the time the transaction takes place.

2. Identify the following statement as true or false: The legal protection regulations on securities and commodities are generally applicable to digital assets.

Hide Solution

Solution:

False. Whereas legal and regulatory standards for traditional financial instruments are well developed, legally tested, and proven; widely and uniformly enforceable; and often comparable across different jurisdictions and legal systems, comprehensive rules that are specific to digital assets—their formation, trading, and legal standards—are being developed. Additionally, the various digital exchanges on which digital currencies are traded are typically not as heavily regulated as traditional financial markets. Consequently, certain irregular market behaviors, such as price manipulation and outright fraud, are not always prohibited on a digital exchange.

3. The ownership and exchange of digital assets on permissionless networks are usually recorded on a:

A. centralized ledger maintained by a central intermediary.
B. decentralized ledger maintained by a central intermediary.
C. decentralized ledger without any central intermediary.

Hide Solution

Solution:

The correct answer is C. The ownership and exchange of digital assets on permissionless networks are generally recorded on a distributed (i.e., decentralized) digital ledger using cryptography and enhanced algorithms without the use of any central intermediary.

4. Compare and contrast the main types of stablecoins.

Hide Solution

Solution:

The traditional type of stablecoin, or “securitized stablecoin,” is backed by physical collateral of fiat currencies, precious metals, or other financial assets as reserves for every stablecoin issued. Each coin can be traded freely on the open market or redeemed for its dollar value from the issuer by liquidating the collateral. A special example of the stablecoin is the asset-backed token, which maintains price parity with some target asset—for example, the US dollar or gold—through tokenization.

Another unique type of stablecoin is called a “smart stablecoin” or an “algorithmic stablecoin,” because there is no backing of physical assets for the cryptocurrency. Rather, its value is backed by another cryptocurrency or token and is linked to the US dollar “algorithmically” instead of using actual dollar reserves.

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